From Blind Estimation to Frequency-Domain Control of a Brain–Machine Hand Interface

GG4 — Neural Data Analysis with Adaptive State Estimation | yhl63

Abstract

This report presents the evolution of our approach to modelling and controlling an unknown brain–machine interface (BMI), progressing from a blind linear–Gaussian estimation framework to a frequency-domain control paradigm informed by system structure and empirical observations. Analysis of the BMI decoder and empirical frequency response revealed that the brain's best linear approximation is well-described by a first-order low-pass response, even though a substantial nonlinear component remains (input–output coherence ∼0.6); this explains the poor high-frequency model fidelity and motivates a shift away from time-domain control. The controller exploits only the coarse property that a tone at one frequency yields a latent oscillation at the same frequency, and closes a position loop to correct the rest, making it robust to the nonlinearity it cannot capture. Early LQG/LQI approaches, based on next-step prediction and standard LQR objectives, proved ineffective as they do not account for the system's reliance on oscillatory inputs. Exploiting linearity and stability, we reformulate control in the frequency domain, enabling open-loop frequency matching and eliminating the need to track fast neural dynamics. By leveraging time-scale separation, we close a single outer loop on position using steady-state responses. The resulting controller achieves consistent performance within 6 cm, with limitations directly attributable to low-pass attenuation at higher frequencies.

1 Problem Definition and Introduction

The system is composed of three principal modules. The neural dynamics from a "brain" are estimated as a linear Gaussian state-space model (the assumptions of this model will be discussed later), which maps a two-dimensional input signal $u \in [0,1]^2$ to a 16-dimensional neural activity vector $y \in \mathbb{R}^{16}$. A brain–machine interface (BMI) then projects y onto a two-dimensional latent state (x_1, x_2) via a linear transformation. This latent representation serves as input to a recurrent neural network (RNN) that produces muscle activation signals for a biomechanical arm model comprising planar shoulder and elbow joints actuated by four antagonistic muscles.

$$u \rightarrow \underbrace{\text{brain}}_{\text{LGSSM}} \rightarrow y \in \mathbb{R}^{16} \rightarrow \underbrace{\text{decoder}}_{(x_1,x_2)} \rightarrow \underbrace{\text{muscle head}}_{\text{nonlinear}} \rightarrow 4 \text{ muscles} \rightarrow \underbrace{\text{2-link arm}}_{\text{integrator}} \rightarrow \text{hand},$$

Our final objective is to steer a simulated hand to a Cartesian target using a closed-loop controller. Our approach is based on a sequence of insights and design choices, motivated by the structure of the BMI and the empirical data obtained through the Brain model. We explore a series of exploration approaches following our interim report and the insights they provided, leading to our final controller design and its evaluation.

2 From blind estimation to active probing

Week-2 starting point. The interim work solved a more difficult estimation problem by attempting to recover both the latent state and an unknown input from observations alone, with no access to u. We modelled the brain as a linear-Gaussian state-space model and closed the problem with an augmented-state construction: the input is stacked onto the latent, $z_k = [x_k^\top \; u_k^\top]^\top$, under an AR(1) random-walk prior $u_{k+1} = u_k + \eta_k$, giving the block-triangular dynamics $z_{k+1} = \left[\begin{smallmatrix} A & B \\ 0 & I \end{smallmatrix}\right] z_k + \tilde{w}_k$, $y_k = [C \; D] z_k + v_k$, such that a single Kalman/RTS smoothing pass returns the state in its leading n and the input in its trailing m coordinates (where n and m are the dimensions of the latent and input spaces, respectively), while EM (closed-form M-step) fits the matrices. We warm-started the EM from PCA of the observations (C from the leading n singular vectors, A from one-step least-squares on the PCA scores), with a few random restarts against local optima. We discussed the limitations of folding the input into the latent : the AR(1) increment prior biases the recovered input toward smoothly-varying signals and represents sharp or impulsive inputs only approximately, and the state/input split is identified only relative to that prior (and the exogeneity constraint of the zero lower-left block)

Known inputs with a black-box model. In Weeks 3–4 we command u, so identification becomes well-posed and enables probe design. Under amplitude constraints ($[0, 1]$ clipping), the most informative inputs lie at the bounds: we employ multi-level, piecewise-constant signals with holds across timescales $1, 4, 16, 64$ as long holds concentrate energy at low frequencies to excite slow dynamics within finite data, while short holds probe faster modes; multiple amplitude levels also expose nonlinearities. This approach outperforms impulses (energy-limited under clipping) and Gaussian noise (high crest factor), and aligns with D-optimal design by maximising $\det M$. For the observed low-pass system, concentrating power in the low-frequency band where control authority is needed yields the most informative data.

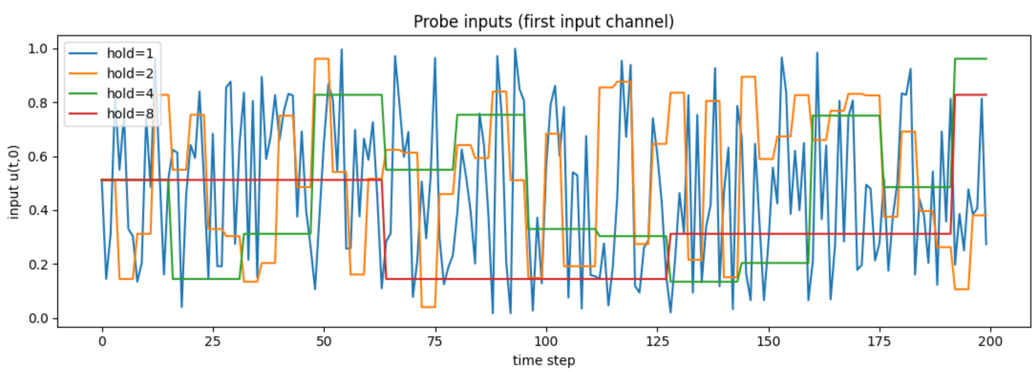

Figure 1: The persistently-exciting probe: multi-level, piecewise-constant signals with holds across timescales.

Knowing the inputs removes the need for the input prior and the augmentation, allowing us to improve on our previous blind estimation by seeding EM with an input-aware subspace estimate via the block-Hankel matrix. The leading left singular vectors of a block-Hankel matrix of the output data, after projecting out the future-input dependence (a MOESP-style step), give a consistent estimate of the extended observability matrix, from which C and A follow by shift-invariance. With (A, C) fixed, (B, x_0) follow from a linear least-squares regression of the observed outputs onto the simulated responses to unit impulses in each input channel and unit initial-state directions. This data-driven warm start, enabled by the persistently-exciting probe, seeds B (which PCA cannot) and remains usable when the number of latent variables exceeds the number of outputs $n¿p$ (which PCA also cannot), placing EM in a markedly better basin. An ambiguity remains, however, in the choice of Hankel window-length, which we determine based on the number of latent dimensions we choose to fit and the number of held-out predictive score.

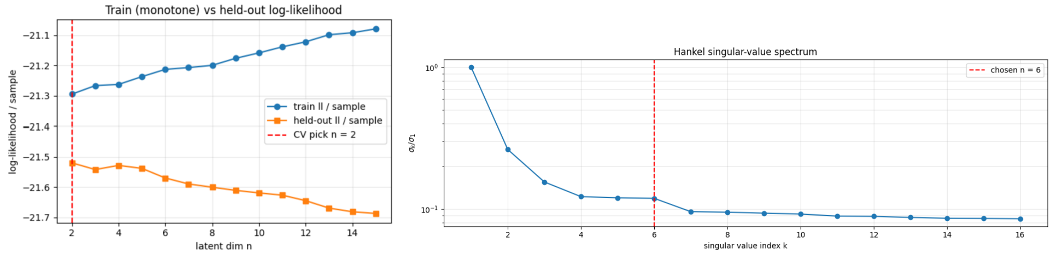

Figure 2: Model order selection diagnostics. Held-out predictive score (left) and singular-value spectrum of the block-Hankel matrix (right) both suggest retaining $n = 6$ modes, with a clear gap after $n = 2$ and a noise floor from $n \approx 6$ onward.

Held-out predictive scoring, repeated over independent simulated trials rather than data-recycling cross-validation, places the model order on a plateau at n=2, consistent with m=2 inputs ($\leq$ 2 controllable directions) and a 2-D decoder readout. From observing the trained-log-liklihood (in blue) we see that the model performs well on the training data but begins to overfit as the number of modes increases, while the held-out log-likelihood (in orange) decays after n=2. The same block-Hankel matrix offers a complementary, structural check via its singular-value spectrum: σ_k/σ_1 drops sharply to k=2, declines gradually through k≈3–6, and flattens into a noise floor from k≈6 onward, the soft-knee signature expected under measurement noise, where the floor marks directions blending into noise. The two diagnostics ask different questions: the spectrum asks whether a mode is distinguishable from noise in this probe at all, while the held-out score asks whether estimating that mode's parameters from this much data improves free-run prediction. We read the gap between n=2 and the n≈6 floor as weakly-excited modes that are real but not reliably identifiable from a probe of this length, and retain n=6 as the operating order — trading predictive parsimony for the controllability margin the downstream LQR/LQI design relies on.

3 The BMI structure forces a frequency-domain view

Inspecting the interface reframed our approach. The decoder taking the 16-D neural activity y to the 2-D latent state z is affine, so it drops into any linear controller unchanged. The muscle head, however, is nonlinear, stateful, and rhythm-coded: muscle selection is the band-energy of x_1's recent history through four fixed band-pass filters, in other words, the oscillation frequency of x_1 chooses the muscle while power gates on x_2 clearing a threshold with persistent evidence. Activation is power × selector, and the arm integrates muscle differentials into joint angular velocities, with a dry-friction deadzone. As a result, we find that a constant x_1 produces no band energy and hence zero selection, to move the hand, x_1 must oscillate, leading to the interpretation that the natural language for the inner control problem is the brain's frequency response. This is what turned the investigation from time-domain regulation toward frequency matching.

4 System identification and why it struggles

We measured the brain's frequency response empirically with a periodic multisine (averaging periods for a free noise estimate) and compared it to the identified model (Fig. 3). Three findings shape the controller.

(i) The model is faithful only in a low band and one direction. With m=2 the response has two principal gains σ_1, σ_2. The dominant σ_1 matches at low frequency, but σ_2 — the second channel of controllable authority — is under-represented by the model almost everywhere, and the relative reconstruction error climbs toward 1 at high frequency where the brain transmits little. **(ii)**

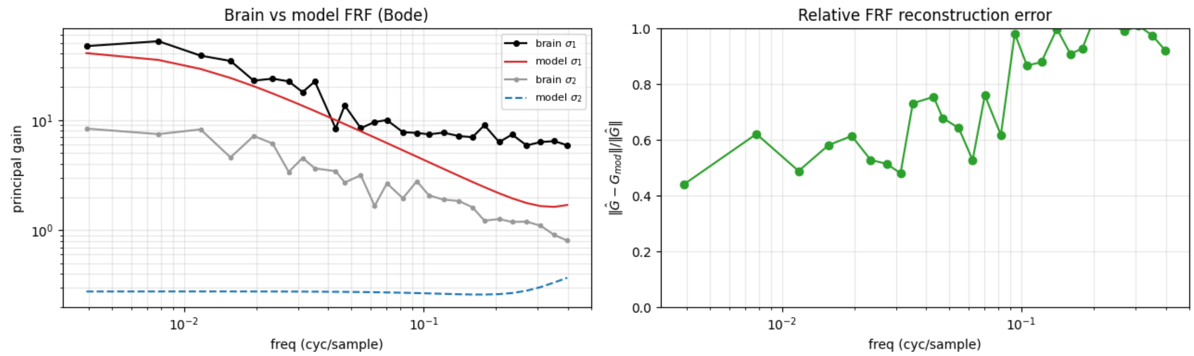

Figure 3: Empirical (black/grey) vs. identified-model (red/blue) principal gains (left) and relative reconstruction error (right). σ_1 is captured at low frequency; σ_2 is badly under-modelled and the error grows with frequency, where coherence collapses — the regime the controller must avoid.

A coherence ceiling reveals nonlinearity. Input–output coherence never approaches 1 even at best SNR: a fixed fraction of the output is not *linearly* explainable, the fingerprint of the nonlinear stack, and no increase in latent dimension removes it (averaging lifts the high-frequency, SNR-limited deficit but not the low-frequency ceiling). **(iii) The model is a linearisation.** Matching frequency responses identifies the system precisely because complex exponentials are the eigenfunctions of an LTI map and $H(z)$ is the similarity-invariant object; against a *nonlinear* plant it returns the best linear approximation at the probed amplitude — which is exactly what a linear controller sees, and is adequate only where coherence is high. These are honest limitations: our linear surrogate is a low-frequency, dominant-direction approximation, and the controller is designed to live within that band.

That frequency matching identifies the plant at all rests on one fact: complex exponentials are the eigenfunctions of an LTI map, so $Y(e^{j\omega}) = H(e^{j\omega})U(e^{j\omega})$ and H on the unit circle *is* the input–output map (its inverse transform is the impulse response). It is also the *right* object because, under the gauge $x \to Tx$, the transforms cancel in $C(zI - A)^{-1}B$: internal coordinates are never identifiable, but H is. So the EM-fitted model and the model-free empirical transfer function *should* coincide, and where they do not (Fig. 3) the gap is genuine model error rather than an artefact of coordinates — which is why the FRF is the diagnostic we trust.

5 The brain is a first-order low-pass system

Across seeds the identified A has a *single dominant mode*: a pole pair at $|\lambda| \approx 0.93$–0.95 with all other eigenvalues much smaller. Fitting the dominant gain $\sigma_1(f)$ to a first-order low-pass $|k/(1 - ae^{-j\omega})|$ gives $a = 0.940$ and $R^2 = 0.998$ (Fig. 4), a $23\times$ roll-off with a -3dB cutoff near $f_c \approx 0.012$ cyc/sample. *The brain is, to excellent approximation, a first-order low-pass filter* in its dominant input–output direction. This *confirms, with inputs, what the blind phase already suggested*: in Weeks 1–2, before any input could be commanded, the observation power spectrum was low-frequency-dominated with no narrow-band peaks — the signature, noted then, of a stable low-order LG-SSM driven by white noise — and PCA placed a low ceiling on the observation rank. The active multisine sharpens that qualitative reading into a quantitative first-order roll-off, so the two phases are consistent and the low-pass character is not an artefact of probing.

This is the pivot of the whole design, for two reasons. First, *linearity in the inner channel* ($u \to z$, since the decoder is affine) means the latent response to a tone is exactly the frequency response at that tone — we can synthesise the oscillations the muscle head demands by *frequency matching*, in open loop. Second, the muscle selection/power frequencies ($0.073, 0.133, 0.207, 0.312$ cyc/sample, §7) all lie *well above* f_c, on the roll-off; so every muscle is driven through attenuation, and the higher-frequency

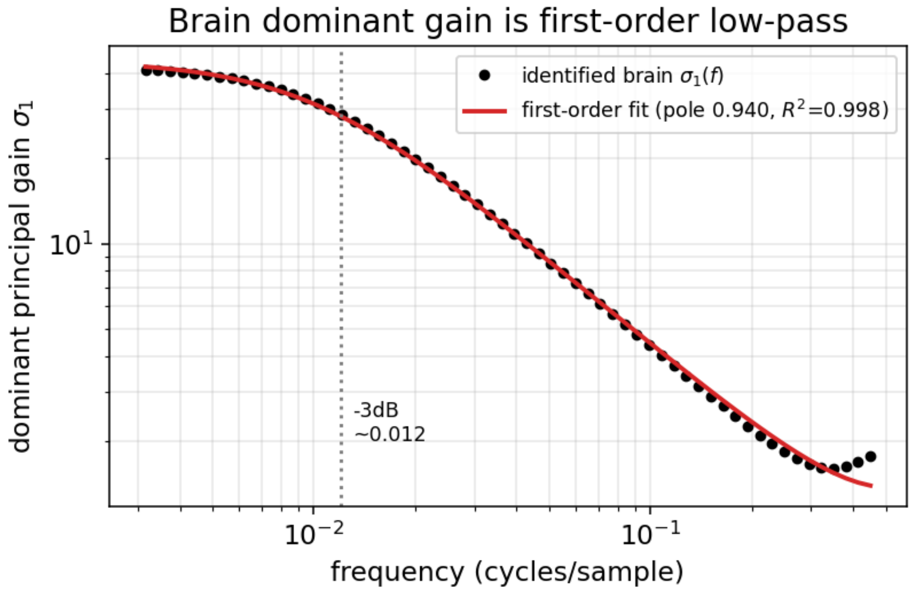

Figure 4: The identified brain's dominant gain $\sigma_1(f)$ (markers) against a first-order low-pass fit (red, R^2=0.998, pole 0.94). The muscle frequencies sit far above the -3dB cutoff, on the roll-off.

muscles progressively more so. This single mechanism predicts the controller's dominant failure (§8) before any controller is built.

6 Initial control: setpoint LQG/LQI, and why it changes

Our first controllers were the standard model-based ones, built on the identified plant. Setpoint regulation chooses a feasible target $u^\star = G^+ y^\star$, $x^\star = (I - A)^{-1} B u^\star$ (G is 16×2, so $y^\star$ is met only in least squares), and applies the *steady-state* LQG law

$$u_t = u^\star - K(\hat{x}_t - x^\star),$$

with K from the control algebraic Riccati equation and $\hat{x}$ from a Kalman filter (LQG = LQR + estimator by the separation principle; the two "Q, R" — cost weights versus EM noise covariances — are distinct). The feedforward $u^\star$ is essential: dropping it drives $u \to 0$ at steady state and the setpoint is not held; a residual offset remains because $u^\star$ uses the *model's* gain, which **LQI** removes by integral action on the tracked output. The Kalman correction makes this a *next-step-prediction* scheme: the estimator re-anchors the state each step from the innovation $y - C\hat{x}$.

Under-actuation shapes what any such law can do. With m=2 inputs and p=16 outputs the steady-state reachable set is $\operatorname{im} G$ (rank ≤ 2), so an arbitrary 16-D output is unattainable and a target is met only in its least-squares projection $u^\star = G^+ y^\star$. The natural controlled variable is therefore a *tracked output* $z = Wy$ of dimension $\leq m$ — and the affine decoder makes the meaningful choice $z = (x_1, x_2)$, the two latent rhythms, available for free. The latent transfer function $H_z(e^{j\omega}) = WC(e^{j\omega} I - A)^{-1} B$ is then a 2×2 map, trivial to invert — the object the frequency synthesis of the next section uses.

Two corrections to our initial reasoning matter. The LQR horizon is *not* the experiment length: K is the infinite-horizon steady-state gain (the algebraic Riccati solution), legitimate because a stabilising loop's time-varying gain converges to it. And minimising $\sum (x - x^\star)^\top Q (x - x^\star) + u^\top R u$ regulates toward a *constant* $x^\star$; to instead *track an oscillation* the cost gains a reference-carrying term and the optimal law becomes $u_t = u_t^\star - K(x_t - x_t^\star)$, whose feedforward $u_t^\star$ for a single tone is exactly the frequency-response inverse $U = H_z(e^{j\omega})^+ Z$ — the Riccati recursion is unchanged (so the problem stays well-posed and convex), only the linear costate term moves. In other words, LQ-tracking of a sinusoid *is* frequency matching.

The decisive issue is structural, not implementational: every setpoint scheme holds the latents *constant*, which (§3) yields *zero* muscle activation. Setpoint LQG/LQI is the right tool for "hold the latents at a level" and the wrong tool for "make them a rhythm." This is what pushed us from regulation to the frequency-domain approach.

7 The final controller: frequency-multiplexed joint servo

The brain being a stable, linear, low-pass system lets us replace the inner regulation problem with open-loop *frequency synthesis*, and — the key simplification — lets the only feedback live at the outer, position level.

Why steady-state responses suffice (no constant neural observation). The brain's dominant pole 0.94 gives a transient time constant of $\tau \approx 1/(1-0.94) \approx 17$ steps and a settling time of a few τ ($\sim$50 steps), *far* shorter than the control horizon (T=1000) and the $\sim$tens-of-steps cadence of a hand correction. So after a brief transient the latent oscillation is at its *steady state*, fully described by the frequency response; we need only place the right spectral content and read the *slow* consequence (the hand), never the fast neural transient. This is precisely what licenses a single *outer* loop on the measurable hand position and an open-loop inner synthesis: the outer loop, operating slower than the inner transients, launders the inner loop's model error and stabilises the velocity$\to$position integrator.

Inner synthesis: a ladder to "free power". The inner loop was built in three stages, each fixing the previous one's failure — the same constructive style as the estimation ladder of §2. *(i) Naive two-tone.* Drive x_1 with a select tone (to choose the muscle) and x_2 with a slow power tone (to power it). This fails: the select tone, transported into both latents, contaminates x_2, which dips through the wake threshold, and because the head's power exits fast but re-enters slowly, each dip collapses power into a single "fire-once" slew that cannot be retriggered. *(ii) Input-direction nulling.* Use the two channels to drive x_1 while *nulling* x_2 at the select frequency (a 2×2 solve on H_z). The nulling itself is well conditioned, but the scheme still needs a *centred* x_2 oscillation, and centring its large offset consumes the $[0, 1]$ headroom, so the select tone clips — and clipping re-introduces the very contamination it removed. *(iii) Free power (final).* A glass-box replica of the head (matching the supplied network to 3×10^{-7}) revealed that x_2's natural DC offset ($\approx +1.8$) already exceeds the wake threshold, so a *constant-high* x_2 sustains power indefinitely. We therefore drop the power tone and the centring entirely and emit only a small select tone — kept small enough that its leakage never dips x_2 below threshold and the input never clips, yet large enough that x_1 clears the selector threshold. The obstacle that sank stage (ii) becomes the feature that powers the controller.

Outer servo. Muscles 0/1 drive the shoulder and 2/3 the elbow, and each joint's velocity depends only on its own pair, so control decouples into two independent inverse-kinematics joint servos. Using the known arm geometry (link lengths — physics, not the network weights), the target and current hand positions are inverted to joint angles; each joint error emits a select tone of amplitude $a = \mathrm{clip}(k|e|, a_{\min}, a_{\max})$ (elbow boosted, total capped to share the budget), and inside a deadzone the tones drop and stiction holds the pose. The four select frequencies and the joint map are *brain-invariant* and calibrated once offline (Fig. 5); the per-brain gain that varies is absorbed by the loop. Two variants differ only in the joint estimate: **V1** reads it from the measured hand by inverse kinematics; **V2** runs a forward observer (the known fixed downstream replayed on the neural) with *no* hand feedback — a Kalman predict-plus-update versus predict-only contrast.

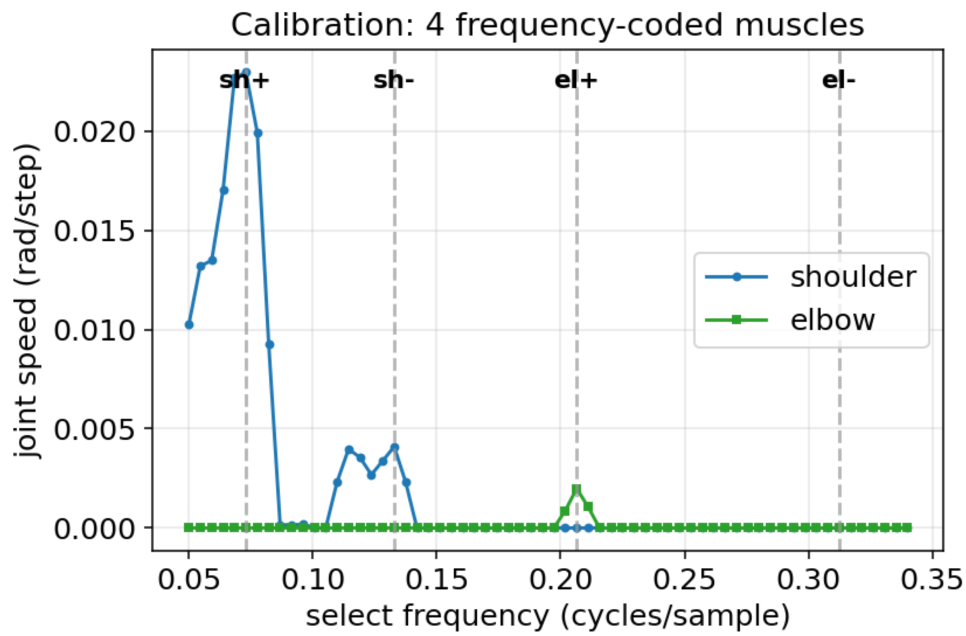

Figure 5: Offline calibration: sweeping the select frequency and measuring steady joint velocity exposes four band-pass muscle peaks. Higher-frequency peaks are weaker — the low-pass roll-off of §5 seen at the actuator level.

8 Evaluation methodology and results

Methodology. All controllers are run through the group-standard harness for commensurability: an identical polar target grid (radii $\{20, 38, 55\}$ cm $\times$ 8 angles), a fixed brain seed, horizon T=1000, and six metrics chosen to expose distinct behaviours — final distance (task error), time-to-threshold (responsiveness), steady-state distance (settling), the *shoulder/elbow* decomposition (which localises failure), control effort (efficiency/saturation), and a spatial heatmap (the *structure* of success). The grid spans the whole workspace because the failure turns out to be directional and a few targets would hide it.

Aggregate. On the grid (Fig. 6) the final distance is 32.3 ± 33.4 cm, shoulder error $19.3 \pm 23.1°$, elbow error $53.1 \pm 63.0°$, time-to-threshold 401 ± 211 steps, effort near-constant (617 ± 23). Every standard deviation rivals its mean: the average hides a sharply *bimodal* outcome.

A directional failure, predicted by the low-pass model. Splitting by workspace half:

half	final dist.	|sh. err|	|el. err|
upper ($y > 0$)	5.7 cm	7°	7°
lower ($y < 0$)	61.6 cm	47°	101°

The upper half is reached to a few cm; the lower half misses by ∼60 cm, almost entirely in elbow error. The two halves recruit different members of each antagonist pair, and measured joint authority falls monotonically with frequency — shoulder+ (0.073) gives 9.7 rad of travel, shoulder− (0.133) 2.0, elbow+ (0.207) 2.0, elbow− (0.312) only 0.9 — exactly the ordering of the low-pass roll-off (§5): the highest-frequency muscle, elbow−, is near-dead, so the lower half is reach-limited. The flat effort across radius (Fig. 6f) confirms an *actuation* limit, not under-driving: more stimulation would not help.

Generalisation and the value of feedback. Calibration is done once but every trial is a new random brain; across five seeds the controller reaches a median 10.2 cm with 93% within 15 cm — the

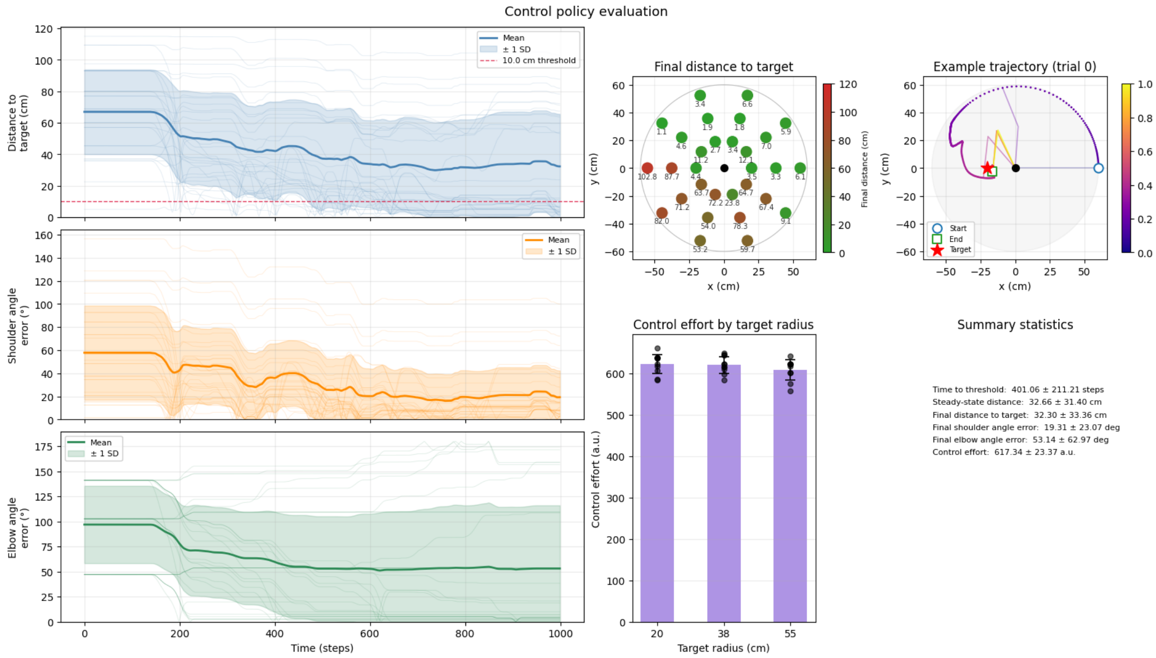

Figure 6: Standardised evaluation (seed 0, $T{=}1000$). (a) Distance: a $\sim$150-step power ramp then a plateau with a wide band (bimodal). (b,c) The shoulder error falls but the *elbow* error stays high. (d) The final-distance heatmap is sharply *directional*: a reachable upper/right region (≤ 12 cm) and a red lower/far-left shortfall (53–103 cm). (e) Example reach. (f) Effort is flat across radius — the failing targets spend effort without converging (an actuation limit).

brain-invariant structure transfers and the loop absorbs each brain's gain. V1 (hand feedback) reaches median 10.2 cm; V2 (neural observer only) reaches 33.7 cm with 0% within 15 cm (Fig. 7): the observer drifts on the $\sim$10$\times$ noisier single-sample neural, so the $\sim$24 cm gap quantifies the worth of direct hand sensing. The two share the entire control law and differ only in the joint estimate, so this is a clean measurement of feedback's value rather than a tuning difference.

A methodological result on tuning. On the calibration brain alone, raising the elbow gain improves accuracy; cross-seed validation *overturns* this — reliability falls from 90% to 65% as it is raised, because over-driving the elbow on brains where it is adequate induces overshoot. Single-brain tuning over-fits; parameters must be chosen by cross-seed reliability. The deadzone, by contrast, is nearly inert.

9 An attempted improvement: muscle-aware kinematic redundancy

The directional failure of §8 is not purely an actuation limit — it is partly self-inflicted. Almost every hand target is reachable in *two* arm configurations, elbow-up ($+|\theta_{\mathrm{el}}|$) and elbow-down ($-|\theta_{\mathrm{el}}|$), and the controller's default chooses the branch nearest the home pose; for the lower workspace that branch demands the weak, high-frequency elbow− muscle. Forcing the *elbow-up* branch instead reaches the same Cartesian point with the strong, low-frequency elbow+ muscle. This is a one-line change to the target inverse kinematics requiring no re-calibration, and its effect is decisive:

target IK branch	overall	lower half	within 15 cm
default (nearest home)	34.7 cm	61.6 cm	46%
elbow-up (strong muscle)	**10.3** cm	**12.7** cm	**79**%

The lower half collapses from 61.6 to 12.7 cm and reliability nearly doubles. This both confirms the

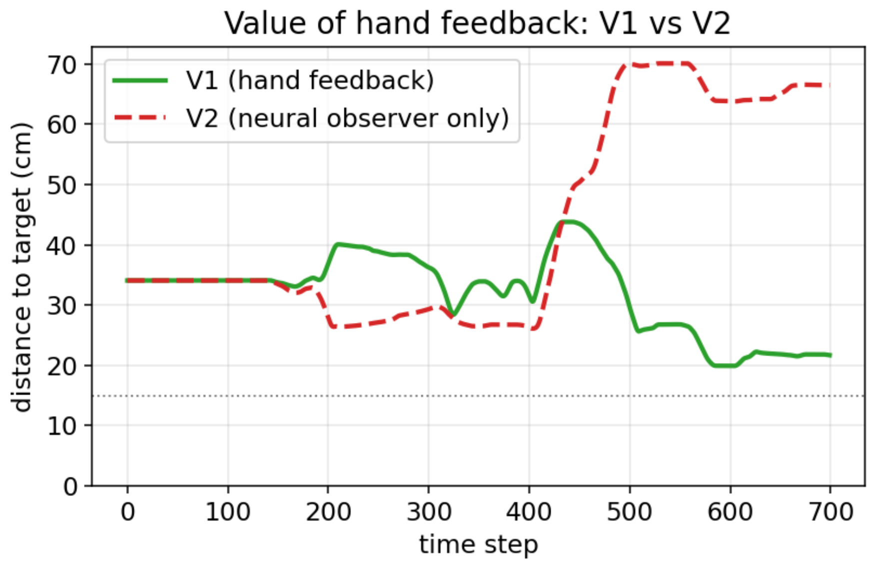

Figure 7: Mean distance-to-target for V1 (hand feedback) and V2 (neural observer, no hand feedback). V1 converges below the 15 cm threshold; V2 plateaus far above it as the forward observer, fed the single noisy neural sample, accumulates joint-angle drift.

diagnosis — the failure was the controller *asking for* an attenuated muscle — and shows that exploiting the arm's kinematic redundancy to route the task onto muscles inside the brain's well-actuated band is the highest-leverage fix. It is the frequency-domain insight applied at the kinematic level. (The default is retained as the baseline reported in §8; the improvement is exposed as a controller flag.)

10 Strengths, limitations, alternatives, and future work

Strengths. The controller exploits rather than fights the decoder (frequency selection, free power), needs no online model inversion, generalises across brains, and its failure is interpretable and traceable to one measured cause.

Limitations. (i) *Elbow under-actuation* from the low-pass roll-off dominates (the directional failure). (ii) The shared $[0, 1]$ budget starves the weak elbow when both joints move. (iii) *Overshoot*: the head's slow power ramp and fast exit make the smallest effective stimulation larger than the final correction, so the hand hunts. (iv) V2 shows the method leans on hand sensing. (v) Steady-state precision is bounded by the friction deadzone.

Alternatives and the 2-DOF synthesis. The rejected schemes each fail a hard requirement: setpoint LQG/LQI cannot produce oscillation; pure Fourier feed-forward cannot cross the nonlinearity or stabilise the integrator; nonlinear MPC needs an honest constrained model of the unknown stack. The coherent picture is a *two-degree-of-freedom* design — frequency-matching feed-forward inside, a single position feedback loop outside — which is what we built; linear MPC would be the natural upgrade to the inner loop to handle the $[0, 1]$ clip honestly. Within the group, a time-domain primitive finite-state-machine controller rests on the *same* four rhythmic primitives but sequences them as discrete bursts; strikingly, both structurally different controllers hit the *same* precision wall, evidence that the binding limit is the plant (power ramp/exit asymmetry and stiction granularity), not the control law.

Future work. Beyond the kinematic-redundancy fix already demonstrated (§9), two improvements remain. A *per-trial elbow self-calibration* (a short probe at trial start) would replace the fixed elbow gain that over-fits one brain (§8); and *sub-primitive actuation* — graded micro-bursts or amplitude annealing on approach — would beat the overshoot wall that no high-level controller removes, the limit shared with the group's time-domain controller.

11 Conclusion

The arc is one of progressive simplification driven by understanding the plant: blind estimation gave way to active, subspace-warm-started identification once inputs were known; the rhythm-coded decoder forced a frequency-domain view; and identifying the brain as a first-order low-pass system collapsed the inner control problem to open-loop frequency matching with a single outer position loop, justified because the controller horizon dwarfs the brain's transient. The result is simple and generalises, and its one dominant limitation — a weak, high-frequency elbow — is not a tuning artefact but the inevitable image of the brain's low-pass roll-off at the actuator. The two transferable lessons are that direct hand sensing is worth $\sim$24 cm here, and that on a stochastic, varying plant rigorous evaluation must be cross-seed.